

Authoring Virtual Peer Interactions for Lecture Videos

Thitaree Tanprasert

tt1996@cs.ubc.ca

University of British Columbia
Vancouver, British Columbia, Canada

Luanne Sinnamon

luanne.sinnamon@ubc.ca

University of British Columbia
Vancouver, British Columbia, Canada

Sidney Fels

ssfels@ece.ubc.ca

University of British Columbia
Vancouver, British Columbia, Canada

Dongwook Yoon

yoon@cs.ubc.ca

University of British Columbia
Vancouver, British Columbia, Canada

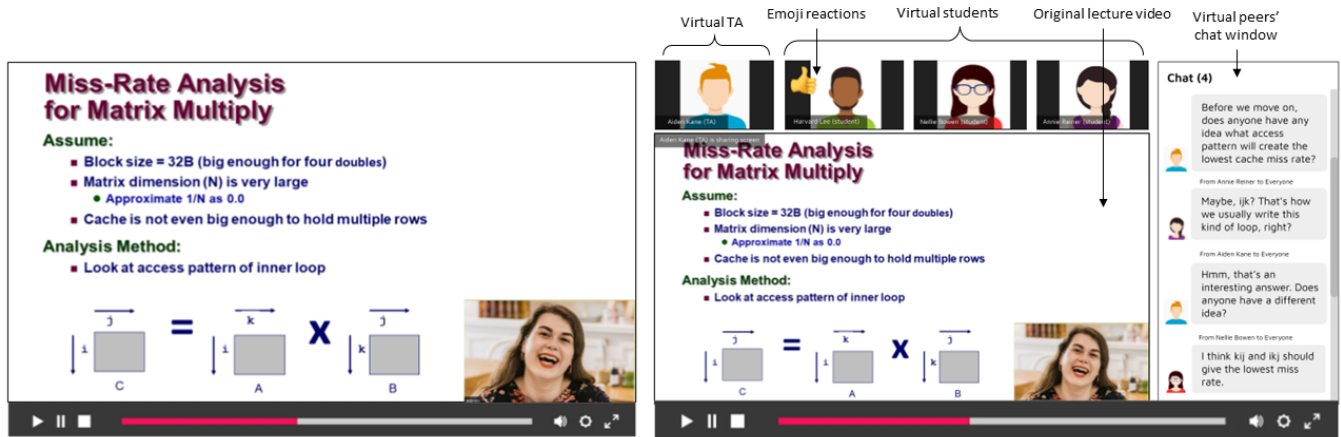


Figure 1: A comparison of the input lecture video (left) and the expected output video from the authoring tool (right). The appearances and interactions of the virtual peers are arranged in the way that mimics Zoom interface.

ABSTRACT

Lecture videos have become a prevalent learning resource, due to the rising popularity of educational video platforms and the necessity for remote learning during the COVID-19 pandemic. However, lecture videos limit feasible pedagogical approaches, since there are no interactions that students would engage with and learn from in synchronous classes. In this study, we proposed virtual peers - virtual TAs and students that interact with each other as though they are watching the video for the first time together - to help real students learn from observing their interactions. We conducted two rounds of interviews and participatory designs with instructors and TAs to design an authoring tool that would help them create engaging virtual characters and author five types of pedagogically valuable interactions. The design received favorable preliminary feedback from instructors. They expect the tool to add valuable

components that are missing from lecture videos with minimal additional effort on their parts.

CCS CONCEPTS

• **Human-centered computing** → *Empirical studies in HCI; Walk-through evaluations.*

KEYWORDS

video-based learning, virtual interactions, video authoring tool

ACM Reference Format:

Thitaree Tanprasert, Sidney Fels, Luanne Sinnamon, and Dongwook Yoon. 2022. Authoring Virtual Peer Interactions for Lecture Videos. In *CHI Conference on Human Factors in Computing Systems Extended Abstracts (CHI '22 Extended Abstracts)*, April 29-May 5, 2022, New Orleans, LA, USA. ACM, New York, NY, USA, 7 pages. <https://doi.org/10.1145/3491101.3519835>

1 INTRODUCTION

Lecture videos have become a prevalent information source for learning, due to the surging popularity of student-centered pedagogy like flipped classrooms, the availability of educational videos platforms like MOOC, and the necessity for remote learning under the constraints of the COVID-19 pandemic [6]. To students, it enables personal learning environment by allowing each student to access at their own convenience. An instructor can also produce

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

CHI '22 Extended Abstracts, April 29-May 5, 2022, New Orleans, LA, USA

© 2022 Copyright held by the owner/author(s). Publication rights licensed to ACM.

ACM ISBN 978-1-4503-9156-6/22/04...\$15.00

<https://doi.org/10.1145/3491101.3519835>

such videos alone, using basic video conferencing or video editing tools to achieve the acceptable quality. Moreover, the video can be re-edited for content update or reused for multiple offering of a course.

Despite the benefits, there are pedagogical limitations of lecture videos. Firstly, it features the instructor alone and does not have any interpersonal interactions, such as questions from other students or peer discussion, which students may learn from interactive, synchronous classrooms (both in online and offline settings) [2]. Moreover, research shows that students may feel isolated, which affects their motivation to learn [12]. These limitations provide an opportunity for enriching the learning environment around lecture videos.

In this study, we proposed a novel solution: adding *simulated* interpersonal interactions between virtual peers into lecture videos. The virtual peers refer to characters with roles in classrooms, such as teaching assistants (TAs) and students. Their interactions are authored under the context that they are watching the lecture video for the first time together on a video conferencing platform (e.g., Zoom) and engage in activities and discussions related to the video along the way. Based on the Active Viewing framework for video-based learning, the virtual peers can act as a proxy of interactive and active video-based learning behaviors [9]. We hope that they can help students learn contents not only from the lecture video itself but also from observing the interactions of the virtual peers around the video. (See Figure 1(right).)

However, to create virtual peer interaction, instructors have to perform many steps of video editing and takes a lot of effort and time. To simplify the authoring process and allow more instructors to enhance their lecture videos, we aim to design a tool will only require raw information from instructors (e.g., a list of frequently asked questions and answers) and automatically turn the information into "realistic" conversation between virtual peers using natural language processing techniques, so that the instructors can create engaging contents with minimal effort.

To design the virtual peer interaction authoring tool, We conducted a participatory design study with 9 participants, who are active university instructors or teaching assistants with experiences in making and/or using lecture videos in their classes. The study aims to answer the following research questions:

- **RQ1.** What types of classroom interpersonal interactions should be simulated for lecture videos?
- **RQ2.** What characteristics should virtual peers have to generate beneficial student engagements?
- **RQ3.** How can we design the tool for authoring the virtual peer interactions into existing lecture videos?

Section 2 of this paper explains the related work and how this work builds upon them. Section 3 illustrates the details of the participatory design study method. Section 4 describes the final design generated from the study and its preliminary feedback. Section 5 concludes the study and discusses the future direction of this project.

2 RELATED WORK

2.1 Adding social components to video-watching experience

There are many approaches to adding social components to make a video-watching experience more enjoyable. In educational settings, there have been attempts to use empathetic agents and Danmaku comments. They range from an interface that motivate students through texts and animated facial expressions when prompted [4] to a robot that interprets student's actions and adaptively encourage their participation in group learning [1]. The problem with empathetic agents is they are often designed specific to certain purpose or subject, therefore they can be difficult for instructors to use and costly for students. Moreover, there has been an ethical concern surrounding empathetic pedagogical agents regarding data privacy, affective privacy, as well as the implication of virtual relationships between humans and AI [14]. On the other hands, Danmaku comments refer to timestamped, textual comments collected from all previous viewers of the video which fly across the screen. Back-and-forth comments also facilitate instructor-student and student-student interactions for utilitarian purposes [3], while hedonic and social comments help decrease student's loneliness [10]. However, they are also shown to distract students from lecture and require higher cognitive loads [20].

In noneducational settings, reaction videos and co-watching sessions are common ways to add social presence and interactions to video watching. Reaction videos (including memes and GIFs reactions) works through affective metawitnessing, where seeing other people's affective reactions can induce an impression of proximity for spatially distant audiences [15]. Reaction videos can also be costly to make since it requires real people to watch the content and record themselves in advance. Moreover, there is risk of affective deviance, where the incongruence between the reactor's and the audience's reaction to the content can cause negative effects on the experience [18]. Finally, in co-watching sessions, chatting during intermissions have been shown make the experience more enjoyable and engaging [19]. Danmaku comments can also be used in co-watching sessions, although they may present visual clutter [5]. The problem with co-watching sessions is that it requires multiple audiences need to attend a session synchronously, which defeats the purpose of lecture videos that allow each student to access it at their convenience.

In contrast to these solutions, the virtual peers, as well as the authoring tool that we present in Section 4.2, will allow instructors to enhance the lecture video on their own with minimum cost and effort. The instructors have controls over all the interactions and can make sure that they are appropriate and helpful to the students. On the student's side, the enhanced lecture videos remain accessible any time. Moreover, since the virtual peers do not interact with the actual students, the solution circumvents the ethical concerns of intelligent agents.

2.2 Vicarious learning in HCI

Vicarious learning is a concept of learning by proxy of watching other people learning [17]. By intentionally presenting deep or relatable or common questions, students can avoid misconceptions

and gain deeper insights into the knowledge [7]. The important component of creating a vicarious learning experience is to generate the actions and conversations of the direct learners, since it is not a part of the original content from the instructor or the final student. An example of systems that aids in this task is YouTute, which uses online social networking between multiple learners to create vicarious learning contents [13]. In contrast to this, our tool will take inputs directly from instructors, so that the instructors can have complete controls over the content and intentionally create a learning environment that is most effective for their classes.

On the authoring tool for creating vicarious learning experience, Nugraha et al. proposed a tool that adds a tutee agent to act out dialogues with instructor [11]. With this tool, the instructor can annotate how the tutee should behave and film themselves responding to the agent. Building on this, our tool will include the affective and social interactions and interactions between multiple characters (e.g., small discussions, quiz from instructor with various possible answers). We will also attempt to decrease the instructor's workloads and explore the possibility of using machine learning techniques to help with the authoring process.

3 METHOD

We conducted two rounds of participatory design study, similarly to the method used to design toolkits for instructors in Scheepmaker et al.'s paper [16]. The purpose of the first round was to gather information about the interpersonal interactions in actual classrooms that should be included and to learn about potential users' (instructors, TAs, or video creators) concerns, constraints, and requirements regarding the idea of authoring virtual interactions. The session consists of an interview about their teaching and video making experience, a brainstorming activity for interactions they have seen in class, and a collaborative design activity (between 1 participant and 1 research). In the brainstorming activity, the researcher prompted instructors to list as many interactions they had seen in classrooms (both online and offline) as possible. Then, the instructors explained the importance of each interaction to the researcher and brainstormed how they thought the interaction may be replicated for students who watched the lecture video of the class. In the design activity, we prepared design sketches of the authoring tool based on literature reviews (described in Section 2) to prompt the participants to criticize or build upon them.

After the first round of the study, we extracted a set of design guidelines for the authoring tool and developed a design for the authoring tool based on them. Then, we conducted the second round of study to get reviews and feedbacks on the design from potential users. The session consists of a walkthrough of the design sketch, which are visualized on Google Slides and are of higher-fidelity compared to the sketches from the first round. We interviewed participants on how the tools adhere to the design guidelines we synthesized from the first study, as well as the difficulties they expected to face while using the interface.

We recruited participants by snowball sampling through lab and faculty members with the intention to recruit both instructors and TAs from across multiple disciplines, since we would like to gauge the potential of this tool as cross-disciplinary. The participants in

the first round are 6 university instructors and 3 teaching assistants from a total of 7 different departments across science, social science, and humanities fields. Every participant has taught 1-3 courses asynchronously with lecture videos. 6 out of 9 participants have experiences using video editing software, such as Camtasia, Adobe Premiere, Shotcut, and Apple iMovie to create lecture videos. However, 3 participants (P1, P2, and P4) stated that they simply use raw recordings as lecture videos without any editing. In the second round, 4 instructors (missing P5 and P9) and 3 teaching assistants from the first round of the study participated again. All participants were compensated 80 CAD for their first round participation and another 30 CAD for their second round.

4 RESULTS

4.1 Findings from first-round study

We transcribed and analyzed the data from the interview transcripts, brainstorming sheets, and session notes, using the research questions as a framework for the identification of potential interaction types and design guidelines.

4.1.1 Essential interactions to include in the tool. There are five types of interpersonal interactions that are repeatedly mentioned by the participants as important components of their classes and are replicable in online classroom settings. These are:

- (1) Students asking questions to the instructor or TA. This is an informative interaction that helps students gain more insights into the course content. It is valuable because it is initiated by the students, and so the instructor's response can be tailored to student's needs (this is mentioned by all participants during the brainstorming session). Moreover, other students who are more passive or less confident in raising the questions can benefit from the instructor's response by vicarious learning (P1, P2, P3, P5).
- (2) Instructors asking questions to the students. This includes any informative interaction where the instructor tests the student's knowledge with a quiz or a class poll (mentioned by all participants) or prompts the student's to brainstorm for ideas or examples based on a topic or concept (P4, P7). It is valuable because it pushes the students to verify and deepen their understanding of the class content. Moreover, students can get a sense of their standing in class and "feel reassured if they know their mistake is common" (P2).
- (3) Discussion among students that are relevant to the class contents. This is an informative interaction where students learn from each other. (e.g., students working together in group activities or exchanging their notes during lecture). Compared to the first type of interaction, this is more suitable for simpler questions (e.g., asking for a definition of a term, making comparison with previous lessons). Moreover, this interaction can happen synchronously with the lecture, so it does not disrupt students who do not need it (P2, P3).
- (4) Pre-class activities, such as greetings or conversations unrelated to class content. This is a social and affective interaction to "engage the student's attention and bring up the energy of the class" (P9) before diving into the lesson.

- (5) Stickers and emojis. Due to the emoji feature in Zoom, this social and affective interaction appears in synchronous online classroom and has become prevalent during the COVID-19 pandemic. It is valuable because it allows many students to "participate and interact without turning on their camera or microphones" (P5).

4.1.2 Design Requirements of the Tool for Authoring Virtual Peer Interactions. There are three important design guidelines on authoring virtual interactions, including the contents, presentation, and the authoring process:

- **DR1.** *Make the interactions easily noticeable and comprehensible, while not distracting students from the lecture videos.* This is a concern of all participants. P2, in particular, pointed out that, "even when [no interaction] is happening, students may feel like they need to keep an eye out for it all the time, so it might be distracting."
- **DR2.** *Clearly distinguish between the real and virtual components.* The interactions should not deceive students about their standings in the class. For example, if a virtual student always asks intelligent questions, and the real students perceive this as representative of students in the class, they may feel like they are falling behind (P7).
- **DR3.** *Automate as much of the authoring process as possible.* The participants are willing to spend different amount of time on improving their existing videos, with the TAs generally willing to spend more time than instructors. P4 especially emphasized on this guideline: "there's already so much going on... it would be great if [the tool] can just create everything, and all I need to do is review and click 'approve this' or 'reject that'."

4.1.3 Guidelines for Designing the Virtual Peer Characters. There are three important design guidelines for the virtual peers:

- **DG1.** *Provide virtual peers with personalities and backgrounds that engage students.* Participants think it is important for students to be able to "relate to the characters so that they can be more engaged [with the characters' interactions]" (P5). P8 also noted that, beyond their personalities, it may be interesting to give the characters some backgrounds, such as hobbies or relationship with each other, since these are the topics that allow students to discuss and bond over in real classrooms.
- **DG2.** *Create virtual peers with diverse and inclusive identities.* To engage all students, it follows that the characters' identities, such as race and gender, must be diverse and inclusive. However, P4 cautioned that we have to be careful when including other traits, such as proficiency of the course, not to create stereotypes.
- **DG3.** *Prevent the virtual peers from acting or saying anything offensively to the instructor in the video, to each other, or to the students watching.* For instance, P5 pointed out that some instructors feel uncomfortable when students comment on their appearances (e.g., "I like your hair"), so it must not appear in the virtual interactions.

4.2 Authoring tool design

4.2.1 Conceptual design. Based on the metawitnessing and vicarious learning literature, which we already discussed in Section 2, we scoped the virtual peers' interactions to be completely authored and not interactive. That is, the actual students watching cannot interact with anything in the final video and will learn only by observing the interactions of a virtual peer. Consequently, we expected the virtual interaction to increase the learning gain of the students, especially in clarifying misconceptions. Moreover, since all interactions are acted out by the virtual peers and are completely separated from the video, there is a clear distinction between the virtual and real components, which adheres to DR2.

We designed the interactions to mimic a first-time video-watching session between the virtual TA(s) and virtual students. The scenario is that the virtual students are watching the video for the first time, with the virtual TA as the moderator of the session. We chose this scenario because interactions in a co-watching session are naturally done via chat, which is the main channel of communication between students and TAs in Zoom classroom. However, research has shown that chats in co-watching sessions should only be done during intermissions to avoid it being distracting. Therefore, in this scenario, the TA will pause and resume the video when answering questions or giving out quiz questions, in order to not overloading students with watching the videos and reading chats at the same time (DR1).

4.2.2 Virtual peer design. There are two types of virtual peers: TAs and students. For each video, there needs to be at least 1 TA and at least 2 students in order to carry out all types of interactions. The user has to pick the characters before they can start adding interactions.

According to DG1, we decided to include two personality traits for the virtual peers. One is proactive vs. reactive, which influences how often a character initiates a conversation (e.g., asks a question) and can influence student's satisfaction in different ways [8]. The TA can only be proactive by design, but a student can be either. This trait is based on inputs that "some students are more outspoken and some are probably more reserved." (P6, and similar descriptions from P2, P4, and P7.) The other trait is formal vs friendly, which influences the tone of language a character uses in the chat, since it is an obvious personality trait that can be easily observed in just one bubble of chat without context or background, so the students can hopefully detect and engage with them quickly.

Although many instructors would like to have knowledge proficiency level as a character trait (i.e., asking deep vs basic questions), we did not include it in this design because there is a risk of students associating a character's identity with the trait due to the small size of the cast (DG2). Moreover, we did not give the characters any background stories or hobbies, even if they may make the characters more engaging, since the extra conversations may distract students from the lecture content, conflicting with DR1.

4.2.3 Interactions design. There are 5 types of interactions that can be authored with this tool, according to the findings in Section 4.1.1: Q&A, Quiz, Side Discussion, Emoji Reaction, and Attendance. We took care in adhering to DR3 in designing the interface for adding

these interactions by requiring as little user inputs and automating as much of the authoring process as possible, such as turning raw Q&A into natural-sounding conversations. The system also takes care of assigning text bubbles to characters in a way that adhere to their personality traits and time the appearance of the bubble to encourage students to think along and follow the conversation easily. The summary of their descriptions and required user inputs are shown in Table 1.

4.2.4 Tool interface design. The authoring tool interface is designed to mimic the Zoom layout. For the informative interactions, the user has to open a panel by clicking the top menu. Here we show an example of adding one side discussion in Figure 2. The user first picks a template for the type of information they would like to include (for Q&A and Quiz, the template is fixed to be a question and answer(s)) (Figure 2(A)) Then, the user inputs the information into the template (Figure 2(B)) to generate a conversation (Figure 2(C)) and see the overview of all discussions that have been added so far. To emoji reactions and edit the generated chat bubbles, the user can do it in-situ.

4.3 Design reviews

We derived the findings from the interview transcripts and session notes. The findings are categorized according to the design guidelines mentioned above.

4.3.1 The virtual peer interactions will not distract students from the content of the lecture videos. Participants do not expect any of the interactions to be distracting from the content of the lecture videos. In particular, pausing the videos when having a Q&A and a quiz would help reduce the students' cognitive load. However, abruptly pausing videos may cause unexpected disruption, so "there should be a warning or notification sound to inform the students" (P2, and similarly mentioned by all participants except P4 and P8).

4.3.2 There is a clear distinction between virtual and real components of the lecture-watching session. Participants believe that it is important to "tell the students that's it's fake and where [the virtual components] come from" (P8, with every other participant in the second round sharing the same sentiment). Beyond this, participants find the distinction between virtual and real components of the final video to be clear for two reasons: one, there is no change to the lecture video itself, so that real portion is contained area-wise; and two, the design of the avatar appearances "don't give an impression of being real" (P3). One concern is that, with the virtual peers may look too virtual to be engaging. A suggestion from P1 and P2 that may be included in the revision of the design is to make TA characters interact with generated voice instead of the interaction being all in the chat.

4.3.3 The effort required from instructors is minimal and acceptable. Overall, the process of inputting data and selecting timing of the interaction seems efficient enough, since "everything that seems like it can be automated is already automated." (P7) Participants are more worried that predicting questions and common mistakes would be time-consuming. In the future, it would be helpful if the tool can suggest questions or collect data from past audience to help generate the questions (P4).

5 FUTURE WORK

After doing another revision on the design based on suggestions from second-round of the study, we will implement a web-based prototype and conduct user evaluation study to complete the answer for RQ3. For this we will let instructors and TAs from 5-7 different disciplines try using the prototype to improve their lecture videos and evaluate the tool's features and ease of use.

After finishing the user evaluation, our next step would be to identify the pros and cons, as well as the suitable context of usage, of the virtual peer interaction. The research questions for this step are:

- **RQ4.** How does the simulated virtual peer interaction affect the student's learning experience?
- **RQ5.** What types of classes and contents benefit from virtual peer interactions and in what ways?

To answer these questions, we will conduct a large-scale study where we recruit undergraduate students through Amazon Mechanical Turk to watch the videos produced from the user testing. They will also watch lecture videos in baseline learning environment, with equal information but in different presentation formats. For example, the questions and answers in Q&A interactions will be presented as an FAQ sheet attached with the video. After watching each video, we will measure the learning gain by having the students complete a short quiz to test the knowledge learned from the video and accompanying components. They will also complete a questionnaire to evaluate their motivation and satisfactions from their learning experience.

6 CONCLUSION

In this paper, we proposed virtual peers as possible conceptual solution to enhance the learning gain, student's motivation, and satisfaction when learning from lecture videos. We also proposed a design of a tool for authoring and embedding virtual peer's interactions into lecture videos. We first conducted a participatory design study with instructors to learn about the characteristics of interaction and virtual peers that would be beneficial for students. Then, we created a design that received favorable feedback from instructors and TAs. They expect the tool to add valuable components that they have never been able to convey in lecture videos with minimal additional effort on their parts.

ACKNOWLEDGMENTS

This work was partially supported by the NSERC Discovery Grant and CREATE programs, as well as the KEIT grant funded by the Korean government (MOTIE, No. 20009940).

REFERENCES

- [1] Patricia Alves-Oliveira, Pedro Sequeira, Francisco S Melo, Ginevra Castellano, and Ana Paiva. 2019. Empathic robot for group learning: A field study. *ACM Transactions on Human-Robot Interaction (THRI)* 8, 1 (2019), 1–34.
- [2] Mobina Beheshti, Ata Taspolat, Omer Sami Kaya, and Hamza Fatih Sapanca. 2018. Characteristics of Instructional Videos. *World Journal on Educational Technology: Current Issues* 10, 1 (2018), 61–69.
- [3] YANG Bo. 2020. Online Learning Interaction in Danmaku Videos: A Pilot Study. In *EdMedia+ Innovate Learning*. Association for the Advancement of Computing in Education (AACE), 1262–1270.
- [4] Gwo-Dong Chen, Jih-Hsien Lee, Chin-Yeh Wang, Po-Yao Chao, Liang-Yi Li, and Tzung-Yi Lee. 2012. An empathic avatar in a computer-aided learning program

Interaction type	Scenario	User Inputs
Q&A	A student asks a question. The TA pauses the video to answer then resume playing the video.	A question and the corresponding answer
Quiz	A TA pauses the video and asks a question for student(s) to try answering.	A question and its correct answer(s) (optional: possible incorrect answers)
Side Discussion	A student asks a “simple” question (e.g., terminology, link to extra resources) and another student answers it; the video doesn’t pause.	type of information (term definition, external link, etc.) and the information (in template form)
Emoji Reactions	A student uses emojis to respond to the chat conversation to show their feelings or agreement.	Emoji selection, character assignment, and timing of appearance
Attendance	A TA starts the conversation and proactive students respond with greetings.	None

Table 1: 5 types of interactions that can be authored with our tool with the scenarios that they are designed to mimic and the inputs that the users (instructors or TAs) need to provide to the tool.

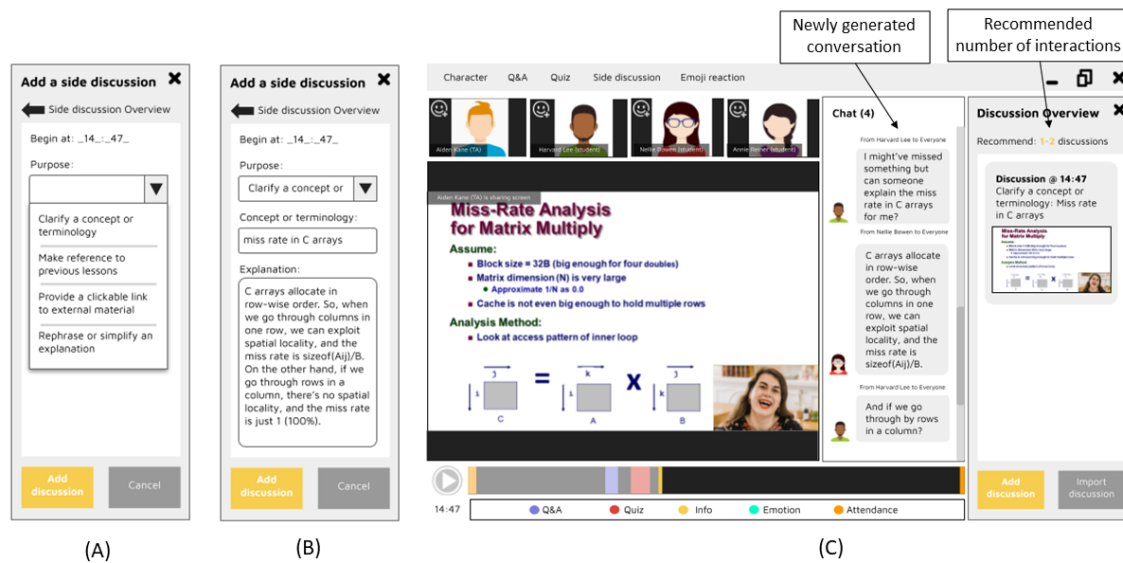


Figure 2: Screenshots of interface for adding a side discussion to a lecture video. (A) selecting the type of information to add; (B) fill out the information in the template; (C) after clicking "Add Discussion" button, the conversation is generated in the chat window. The panel turns into a Discussion Overview window. A yellow bar appears at the timeline below to mark the start of the discussion.

- to encourage and persuade learners. *Journal of Educational Technology & Society* 15, 2 (2012), 62–72.
- [5] Yue Chen, Qin Gao, and Pei-Luen Patrick Rau. 2017. Watching a movie alone yet together: understanding reasons for watching Danmaku videos. *International Journal of Human-Computer Interaction* 33, 9 (2017), 731–743.
 - [6] Konstantinos Chorianopoulos. 2018. A taxonomy of asynchronous instructional video styles. *International Review of Research in Open and Distributed Learning* 19, 1 (2018).
 - [7] Scotty D Craig, Barry Gholson, Joshua K Brittingham, Joah L Williams, and Keith T Shubeck. 2012. Promoting vicarious learning of physics using deep questions with explanations. *Computers & Education* 58, 4 (2012), 1042–1048.
 - [8] Vanessa P Dennen, A Aubteen Darabi, and Linda J Smith. 2007. Instructor-learner interaction in online courses: The relative perceived importance of particular instructor actions on performance and satisfaction. *Distance education* 28, 1 (2007), 65–79.
 - [9] Samuel Dodson, Ido Roll, Matthew Fong, Dongwook Yoon, Negar M Harandi, and Sidney Fels. 2018. An active viewing framework for video-based learning. In *Proceedings of the fifth annual ACM conference on learning at scale*. 1–4.
 - [10] Xi Lin, Mingyu Huang, and Leslie Cordie. 2018. An exploratory study: using Danmaku in online video-based lectures. *Educational Media International* 55, 3 (2018), 273–286.
 - [11] Ari Nugraha, Tomoyuki Harada, Izhar Almizan Wahono, and Tomoo Inoue. 2020. A Tool to Add a Tutee Agent in a Monologue Lecture Video Improves Students' Watching Experience. In *Extended Abstracts of the 2020 CHI Conference on Human Factors in Computing Systems*. 1–7.
 - [12] John O'Donoghue, Gurmak Singh, and Charmaine Green. 2004. A comparison of the advantages and disadvantages of IT based education and the implication upon students. *Digital Education Review* 9 (2004), 63–76.
 - [13] Susen Rabold, Stuart Anderson, John Lee, and Neil Mayo. 2008. YouTube: Online social networking for vicarious learning. In *Proceedings of the ICL2008*. Kassel University Press, 1–14.
 - [14] Deborah Richards and Virginia Dignum. 2019. Supporting and challenging learners through pedagogical agents: Addressing ethical issues through designing for values. *British Journal of Educational Technology* 50, 6 (2019), 2885–2901.
 - [15] Kerstin Schankweiler. 2020. Reaction Images and Metawitnessing. *Parallax* 26, 3 (2020), 254–270.
 - [16] Laura Scheepmaker, Kay Kender, Christopher Frauenberger, and Geraldine Fitzpatrick. 2021. Leaving the field: Designing a socio-material toolkit for teachers to continue to design technology with children. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*. 1–14.

- [17] Leah A Sutton. 2001. The principle of vicarious interaction in computer-mediated communications. *International Journal of Educational Telecommunications* 7, 3 (2001), 223–242.
- [18] Lauren Szczurek, Benoît Monin, and James J Gross. 2012. The stranger effect: The rejection of affective deviants. *Psychological science* 23, 10 (2012), 1105–1111.
- [19] Justin D Weisz, Sara Kiesler, Hui Zhang, Yuqing Ren, Robert E Kraut, and Joseph A Konstan. 2007. Watching together: integrating text chat with video. In *Proceedings of the SIGCHI conference on Human factors in computing systems*. 877–886.
- [20] Yabo Zhang, Aiping Qian, Zhongling Pi, and Jiumin Yang. 2019. Danmaku related to video content facilitates learning. *Journal of Educational Technology Systems* 47, 3 (2019), 359–372.